

Gastrointestinal Endoscopy Classification using Fuzzy Logic and Machine Learning Algorithms

Vasudevan S

Department of Mathematics
Hindustan Institute of Technology and Science.
Padur, Chennai.
svasudevanidvrs@gmail.com

Vediyappan Govindan

Department of Mathematics
Hindustan Institute of Technology and Science
Padur, Chennai.
govindoviya@gmail.com

Abstract— This effort aims to classify lower gastrointestinal pictures using hyperparameter-tuned machine learning classifiers. Lower gastrointestinal images are essential for diagnosing and treating various gastrointestinal diseases. Categorizing these photographs accurately may help identify and plan therapy, improving patient outcomes. SVM, Random Forest, KNN, Logistic Regression, and Gradient Boosting are used classifiers. Each classifier was evaluated using accuracy, classification reports, and confusion matrices. We used Possibilistic C-Means (PCM) for fuzzy classification to increase accuracy. Medical image processing often involves data uncertainty and imprecision, which PCM helps manage. The method gives membership levels to data points for more flexible and accurate categorization. Hyperparameter tweaking was done using GridSearchCV, ensuring classifier performance was optimal. With optimal values for each model, classifier accuracy varied. The Random Forest classifier has the highest accuracy rate of 69.00%, followed by SVM at 68.25%, Gradient Boosting at 68.00%, KNN at 67.38%, and Logistic Regression at 66.38%. Confusion matrices provided a complete assessment of the models' performance. A detailed summary table showed each classifier's ideal parameters and accuracies, revealing their usefulness in classifying lower gastrointestinal images. This study highlights machine learning classifiers' medical image analysis capabilities. It shows the need of fine-tuning hyperparameters and using fuzzy classification methods like PCM for optimum classification results.

Keywords— Fuzzy Logic, Fuzzy Inference, Membership Functions, Python with Machine Learning Algorithms.

I. INTRODUCTION

Human digestive systems are prone to several diseases. Stomach, colorectal, and esophageal cancers cause 2.8 million new cases and 1.8 million deaths annually. Endoscopy is the best GI research method. Physician variation may also occur in endoscopic severity rating and subclassification of results. Among the most common disorders of the digestive tract are gastrointestinal (GI) diseases including polyps, ulcers, and esophagitis; their early identification is absolutely vital to lower morbidity and death. [2] Though video capsule endoscopy and other imaging technologies have allowed non-invasive identification of such conditions, doctors still have a major challenge in deciphering the large volume of GI photographs[4] Consequently, automated categorization systems based on deep learning (DL) and machine learning (ML) have drawn interest in the medical field.

But, from endoscopic pictures, correct categorization of GI tract diseases presents certain natural difficulties. Among these include visual similarities between disease categories

(e.g., polyps vs. esophagitis), image quality variation, noise and artifact presence[5] and the scarcity of annotated datasets, which compromise the performance of traditional ML and DL models[6]. Furthermore, the uncertainty in visual characteristics frequently results in misinterpretation, hence requiring a more strong and smart diagnostic strategy.

By creating a hybrid classification framework combining fuzzy logic-based segmentation, particularly the Possibilistic C-Means (PCM) algorithm, with supervised machine learning classifiers[7] this study seeks to solve these issues. Before classification, the fuzzy logic component is used to improve the interpretability and uncertainty management of the picture features. This method increases diagnostic accuracy, particularly in complicated or visually overlapping GI circumstances, by means of improved region-of-interest (ROI) identification and decreased ambiguity in feature space.

The rationale for incorporating fuzzy logic in this paper originates from its capability to outline uncertainty and manage vague data which is a common attribute in medical image processing, particularly in the imaging of the gastrointestinal tract. Unlike traditional methods of classification that yield distinct results like crisp classification, fuzzy logic offers a degree of membership allowing pixels or portions of an image to belong to multiple classes with different levels of certainty. This is particularly beneficial for differentiating between gastrointestinal diseases such as polyps and ulcers, because the visual characteristics tend to overlap or exhibit subtle differences. Moreover, fuzzy logic increases robustness and reliability because the system can reason and arrive at conclusions in unclear situations, therefore emulating human reasoning. Incorporating fuzzy logic with the Possibilistic C-Means (PCM) method enhances the system's capability to delineate areas of interest and mitigates misclassification due to image noise, illumination changes, and even anatomical differences.

II. REGARDING THE RECORDS

In this part, we examine the Kvasir dataset, which is a collection of images has been curated for the purposes of studying gastrointestinal endoscopy as illustrated in Figure 1. its intended purpose lies within the realms of medical imaging research and education, concentrating on classifying and segmenting different types of diseases of the gastrointestinal tract. Each class in the Kvasir dataset is described as follows.

A. Esophagitis

The esophageal inflammation is the defining feature of this illness. It presents as redness, edema, and ulceration of the esophageal lining that can be seen in the endoscopic photographs. Common reasons include acid reflux, infections, and certain medications.

B. Normal Pylorus

The pylorus is the opening that the stomach join with the duodenum which is the first part of the small intestine. Rest of the organ can properly be appreciated as a rounded and smooth orifice with no vision of any abnormal growth, inflammation or obstruction.

C. Ulcerative Colitis

Chronic inflammatory disease of the colon is ulcerative colitis. Persistent areas of ulceration and inflammation are seen on the colonoscope, the mucosa exhibits hyperemia and also may ulcerate or be purulent.

D. Dyed-lifted-polyps

The common way of achieving polypectomy during an endoscopic procedure is by injecting the mucosal surface polyps with methylene blue or indigo carmine. Making the polyps visible also makes it easier to fully remove them. The dye used enhances the margins of the polyp and other surrounding areas. This aids in the identification of the dissection area which is adjacent to the normal tissue.

E. Dyed-resection-margins

In this excerpt, images with dye meant for viewing the tissue margins where polyps or other lesions have been excised are included. The enhanced visibility provided by the dye helps ascertain unrelated cells have not been retained within the polyp cavity, thus reducing chances of regrowth, and determine that all tissue resection margins are devoid of remaining polyp material.

F. Normal Cecum

The large intestine has the cecum at the beginning, which is the wide pouch like structure. Smooth, healthy mucosal surfaces devoid of tumors, polyps, or inflammation are shown on normal endoscopic pictures. It should be possible to see the distinctive ileocecal valve, which regulates the passage between the small intestine and the cecum, and it should look healthy and functioning

G. Normal Z-line

The squamocolumnar junction, often known as the Z-line, denotes the change in lining from the stomach to the esophagus. Normally, an endoscopic image shows it as a distinct, asymmetrical line. The way it looks is important for recognizing diseases like Barrett's esophagus, where prolonged exposure to acid can cause this line to migrate upward

H. Polyps

During endoscopies, polyps—abnormal growths that emerge from the mucous membrane—are frequently discovered in the stomach or colon. Their appearance might be flat, sessile, or pedunculated lesions, and their size and form can vary. Although the majority are benign, a histological study and excision may be necessary if a polyp is found to be precancerous that's shows in Fig.1

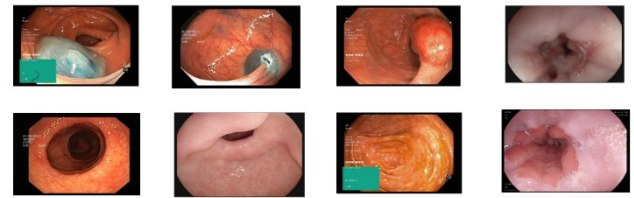


Fig 1. Kvasir Data with Eight Different Classes

III. DEFINITIONS

In these sections the authors discussed a pixel in a picture may have different degrees of belonging to more than one class rather than being rigidly categorized as belonging to one

A. Fuzzy Logic

Unlike typical binary sets, where values are either 0 or 1, fuzzy logic is a type of many-valued logic where truth values are given in degrees between 0 and 1. Mathematically, a fuzzy set of A in a universe X is defined by the membership function $\mu_A: X \rightarrow [0,1]$, where $\mu_A(x)$ represents the degree of membership of x in A. To deal with ambiguities and uncertainties [1], fuzzy logic is employed.

B. Possibilistic C-Means

PCM, also known as Possibilistic C-Means, is a clustering methodology that represents an enhancement over Fuzzy C-Means (FCM). FCM tries to minimize the total of squared errors, but PCM provides a possibilistic approach for more effective management of noise and outliers. PCM, like FCM [2], gives a membership degree to each data point in each cluster. However, in PCM, the membership values are understood as possibilities rather than probabilities

C. Working of PCM

- Initialize the cluster centers and configure the membership matrix's beginning parameters.
- Find the membership values for every data point in each cluster. Unlike FCM, their memberships do not have to sum up to one [3].
- Update the cluster centers based on the weighted average of the data points (weights being the possibilistic memberships).
- Until convergence, iteratively decrease the objective function by updating cluster centers and memberships.

IV. TERMINOLOGIES

In these sections the authors discussed Create the membership matrix, which stores the membership values for each data point in relation to each cluster. These numbers represent the membership degree or likelihood of each data item belonging to each cluster.

A. Specify the initial values for both the cluster centres and the membership matrix.

Cluster Centers: The positions of the cluster centers in the data space are initialized randomly. These centers serve as the initial estimations for the centroids of the clusters [4].

Membership Matrix Initialization: Create the membership matrix, which stores the membership values for each data point in relation to each cluster. These numbers represent the

membership degree or likelihood of each data item belonging to each cluster.

B. Determine the values of membership.

This phase entails calculating the membership values for each data point with respect to each cluster. The membership value, u_{ij} , denotes the level of probability associated with a specific data point x_i belongs to cluster j .

$$u_{ij} = \exp\left(-\frac{\|x_i - v_j\|^2}{\eta_j}\right)$$

Here, $\|x_i - v_j\|$ is the distance between data point x_i and cluster v_j . The parameter η_j controls the spread of the cluster and influences the shape of the membership function [5].

$$v_j = \frac{\sum_{i=1}^n u_{ij}^m x_i}{\sum_{i=1}^n u_{ij}^m}$$

In this context, the variable m represents the fuzzification parameter, which determines the extent to which the membership values affect the clustering process.

C. Calculate the objective function.

Determine the numerical value of the goal function, which the algorithm aims to minimize. The objective function integrates the distances between data points and cluster centers, as well as the possibilistic memberships shown in Figure 2.

$$J = \sum_{j=1}^c \sum_{i=1}^n u_{ij}^m \|x_i - v_j\|^2 + \sum_{j=1}^c \eta_j \sum_{i=1}^n (1 - u_{ij})^m$$

The initial term denotes the cumulative value of the weighted squared distances between the data points and the cluster centers. Meanwhile, the subsequent term discourages low membership values to prevent unimportant solutions.

D. Verify if there is convergence.

Evaluate the convergence of the method by verifying if the alterations in membership values and cluster centres are lower than a pre-established threshold [6].

Convergence criteria: The algorithm is regarded to have converged if the changes in cluster centres and membership values between successive iterations are below a predefined threshold. If the specified conditions are satisfied, proceed to the final "End" step [7]. If the condition is not met, iterate steps 3 to 5 again that represents Figure 2 & 3.

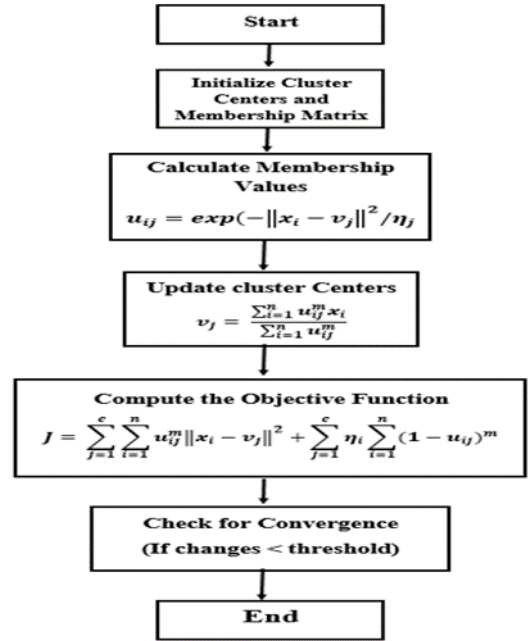


Fig 2. PCM Implementation On Machine Learning

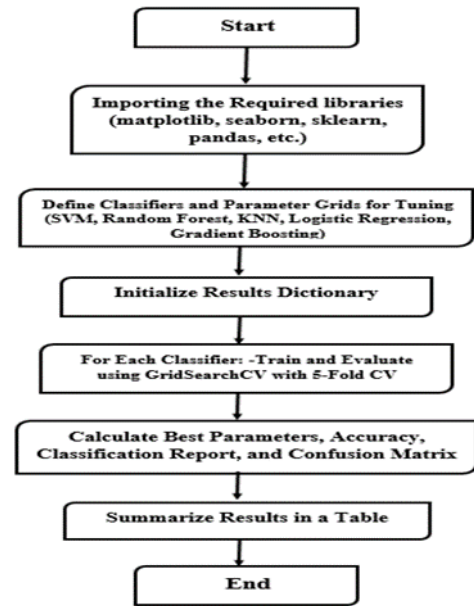


Fig 3. Algorithm Of Possibilistic C-Means

PCM can distinguish ambiguous or overlapping structures in GI pictures more successfully than hard clustering techniques by awarding possibility-based cluster memberships. Cleaner, more concentrated picture areas produced by this segmentation process are then sent into machine learning classifiers—e.g., SVM, Random Forest, Decision Tree—for final categorization. Improved area clarity and lower background noise help the classifiers to pick more pertinent patterns, hence increasing classification accuracy and lowering false positives.

V. EXECUTION

This part provides details of how these various machine learning approaches are integrated into the classification of

lower GI images, focusing on hyperparameter tuning. The implementation uses the Possibilistic C-Means (PCM) fuzzy classification approach detailed in Table 1. To meet the objectives of this research, five classifiers were chosen from several available machine-learning classifiers. Every classifier was trained on the training set that had been modified using PCM shown in Figure 4. GridSearchCV was used to identify the optimal hyper parameters for each classifier. The optimal estimator for each classifier was utilized to make predictions on the test set. The predictions were assessed based on their accuracy, classification reports, and confusion matrices.

TABLE I. HYPERPARAMETER UTILIZED ON PCM CLASSIFIERS

S.No	Classifier	Best Parameters	Accuracy
1	SVM	{'C': 10, 'kernel': 'rbf'}	0.6825
2	Random Forest	{'max_depth': 10, 'n_estimators': 200}	0.6875
3	KNN	{'n_neighbors': 7, 'weights': 'distance'}	0.67375
4	Logistic Regression	{'C': 10}	0.66375
5	Gradient Boosting	{'learning_rate': 0.1, 'n_estimators': 200}	0.685

The specific hyperparameters selected for GridSearchCV tuning—such as the number of estimators and maximum depth for Random Forest, kernel type and regularization parameter C for SVM, and criterion for Decision Tree—were chosen because they directly affect model complexity, decision boundaries, and generalization capability. For instance, the max_depth and n_estimators in Random Forest control how deep and ensemble-driven the trees are, hence affecting overfitting and variance.

The SVM kernel and C parameter decide how the model handles non-linearity and trade-offs between margin maximizing and classification error. These variables were given top importance in tuning to ensure that every model finds the optimal balance between bias and variance, hence improving performance on medical picture data with significant intra-class variation and noise.

Calculating ROC and AUC and creating ROC curves for each. Micro-average and macro-average are used to analyse class performance on ROC curves shown in Figure 4. The ideal parameters, accuracy, classification report, and confusion matrix of each classifier are kept and presented. ROC curves provide exact performance insights, resolve class imbalances with micro-average, and treat all classes fairly with macro-average. This helps understand the pros and cons of each model, guiding model selection and improvement.

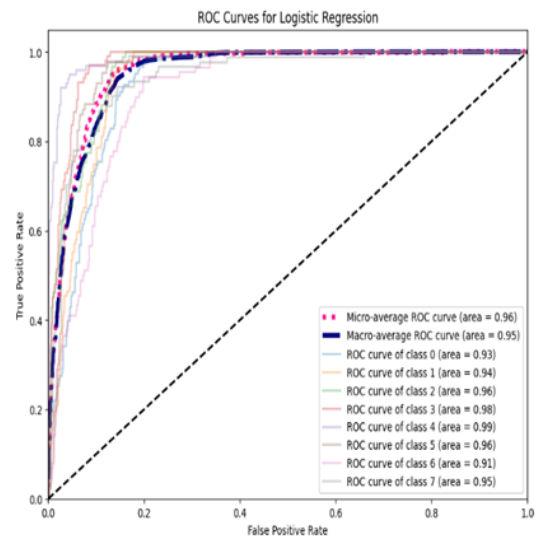


Fig 4. ROC Curves for Logistic Regression

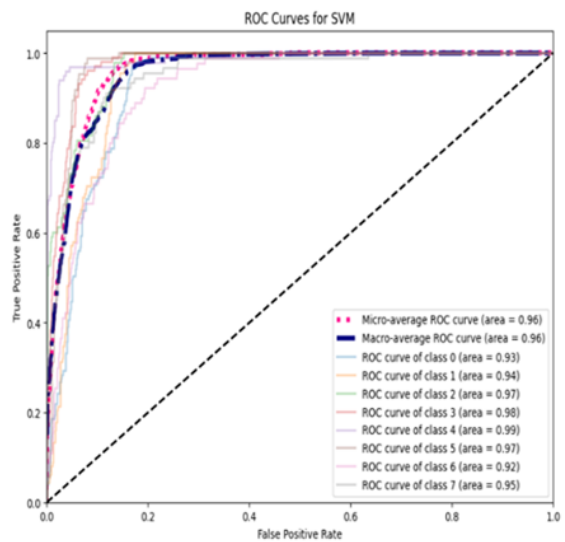


Fig 5. ROC Curves for SVM

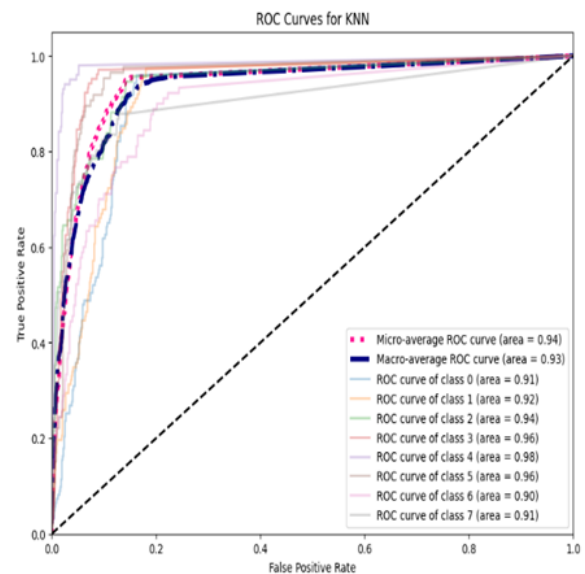


Fig 6. ROC Curves for KNN

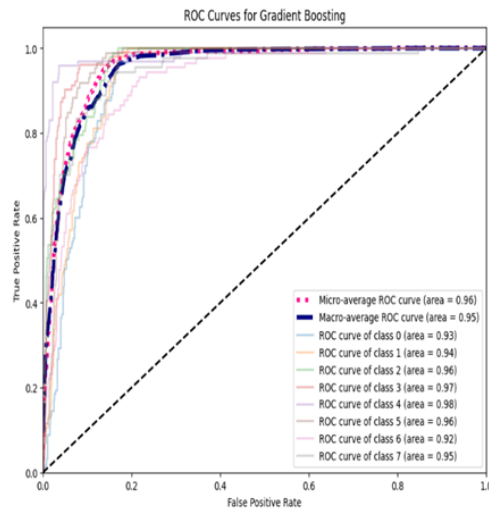


Fig 7. ROC Curves for Gradient Boosting

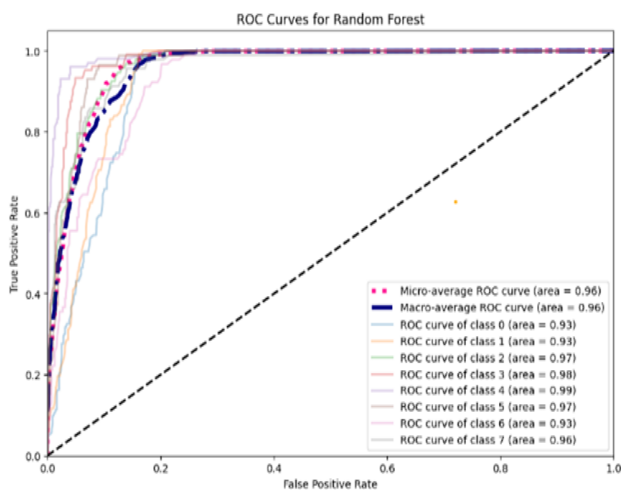


Fig 8. Receiver Operating Characteristic on Machine Learning Algorithms

Besides the ROC curve analysis, we examined the classification reports and confusion matrices to provide a more thorough evaluation of the performance of each model. The classification reports help to assess not only general accuracy but also the models' behavior across imbalanced classes by showing precision, recall, F1-score, and support for each class. Especially for classes 3, 4, and 5, support vector machines (SVM) and random forest classifiers often had high F1s, indicating their capacity to find well-represented and distinctive classes. SVM, for instance, has F1-scores over 0.98 for these categories; but, its performance for class 0 was slightly poorer, likely because to class imbalance or feature overlap. Next was logistic regression, which performed well across all classes but somewhat dropped recall for class 2 and class 6. Gradient boosting showed a similar pattern, offering excellent classification quality for dominant classes but somewhat underperforming on class 7.

K-nearest neighbours (KNN) of all models demonstrated the least consistent performance with much lower F1-scores for class 0 and class 7, suggesting its sensitivity to local data distribution and noise. The confusion matrices verified these results even further by indicating that misclassifications were most frequent between class pairs like class 0 and class 1, and

class 6 and class 7—implying possible feature overlap or visual similarity. Conversely, classes 3 to 5 had minimal misclassification, which reflects their high AUC and F1-scores. Although all classifiers performed well, these comprehensive studies reveal that ensemble methods like random forest and gradient boosting, along with SVM, offer more consistent and fair classification across all classes than logistic regression and KNN.

VI. RESULTS AND CONCLUSION

In Conclusion, Machine learning was used to categorize lower gastrointestinal images in this investigation. From these approaches, the Random Forest classifier has the highest accuracy (69.00%). However, the performance was not sufficient for medical diagnosis. This shows how difficult it is to categorize medical imaging due to its complexity and variety. Data ambiguity was handled using Possibilistic C-Means (PCM) fuzzy classification. PCM provided a framework for fuzzy categorization, but the results showed room for improvement. Complex and sophisticated algorithms are needed to increase medical image categorization accuracy and reliability.

- Being an ensemble technique, Random Forest is able to obtain many decision paths and lower variance making it easier to generalize for more complex visual patterns present in the dataset.
- While SVM kernel-induced boundaries might have some issues with noise-drenched or poorly separated regions of data like in the esophagitis classification, it's a wonderful algorithm for working with a large set of features.
- Decision Tree is easy but overly pragmatic and quick in dealing with data. Because of it being so straightforward, it almost always is getting exposed to overfitting, as most cases of low inter-class variation like those seen in this, it has higher chances of misclassification.

Apart from it being PCM based, the current technique has a remarkable dependency on segmentation that while useful in reducing ambiguity does not in accurately delineating the region of interest in high noise or low contrast scenarios. The dataset, albeit rich in variation, remains considerably small which limits the generalizability of the trained models to broader clinical populations or unseen endoscopic situations.

Research can focus on the augmentation of the dataset, the implementation of deeper neural network architectures and hybrid approaches that fuse fuzzy logic with attention and transformer models to enhance the capture and classification of object and region targets.

Future studies should focus on fuzzy classification techniques like Fuzzy C-means (FCM) and Modified Possibilistic C-Means. Fuzzy C-means (FCM), a clustering technique that allocates data points to many groups with varying degrees of membership, may improve classifications. PCM has limits, while MPCM can manage noise and outliers better. Advanced fuzzy classification approaches may improve lower gastrointestinal imaging classification accuracy and robustness.

Data Availability Statement: The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

Conflicts of Interest: The authors declare no conflict of interest.

Author's contributions: All authors are equally contributed to the research work.

REFERENCES

- [1] S. Kalaivani, C. S. Kanimozhiselvi and V. Rajasekar, "Enhancing the Performance of POS based Features using Generalization for Sentiment Classification," 2022 6th International Conference on Computing Methodologies and Communication (ICCMC), Erode, India, 2022. (*references*)
- [2] D. M. Harikrishna and K. S. Rao, "Classification of children stories in hindi using keywords and POS density," 2015 International Conference on Computer, Communication and Control (IC4), Indore, India, 2015.
- [3] D. M. Harikrishna and K. S. Rao, "Classification of children stories in hindi using keywords and POS density," 2015 International Conference on Computer, Communication and Control (IC4), Indore, India, 2015.
- [4] R. M. Pereira, Y. M. G. Costa, R. L. Aguiar, A. S. Britto, L. E. S. Oliveira and C. N. Silla, "Representation Learning vs. Handcrafted Features for Music Genre Classification," 2019 International Joint Conference on Neural Networks (IJCNN), Budapest, Hungary, 2019.
- [5] C. Leite da Silva, L. May Petry and V. Bogorny, "A Survey and Comparison of Trajectory Classification Methods," 2019 8th Brazilian Conference on Intelligent Systems (BRACIS), Salvador, Brazil, 2019.
- [6] T. F. L. de Medeiros, E. V. C. de Lima Borges, B. É. S. Cavalcante, A. B. Cavalcanti, I. L. P. Andreza and L. V. Batista, "Heart arrhythmia classification using the PPM algorithm," ISSNIP Biosignals and Biorobotics Conference 2011, Vitoria, Brazil, 2011. M. Young, The Technical Writer's Handbook. Mill Valley, CA: University Science, 1989.
- [7] M. Sperle, E. Negri and C. Ternes, "Automatic classification of sidescan sonar images for mapping marine mineral resources," 2015 IEEE/OES Acoustics in Underwater Geosciences Symposium (RIO Acoustics), Rio de Janeiro, Brazil, 2015.